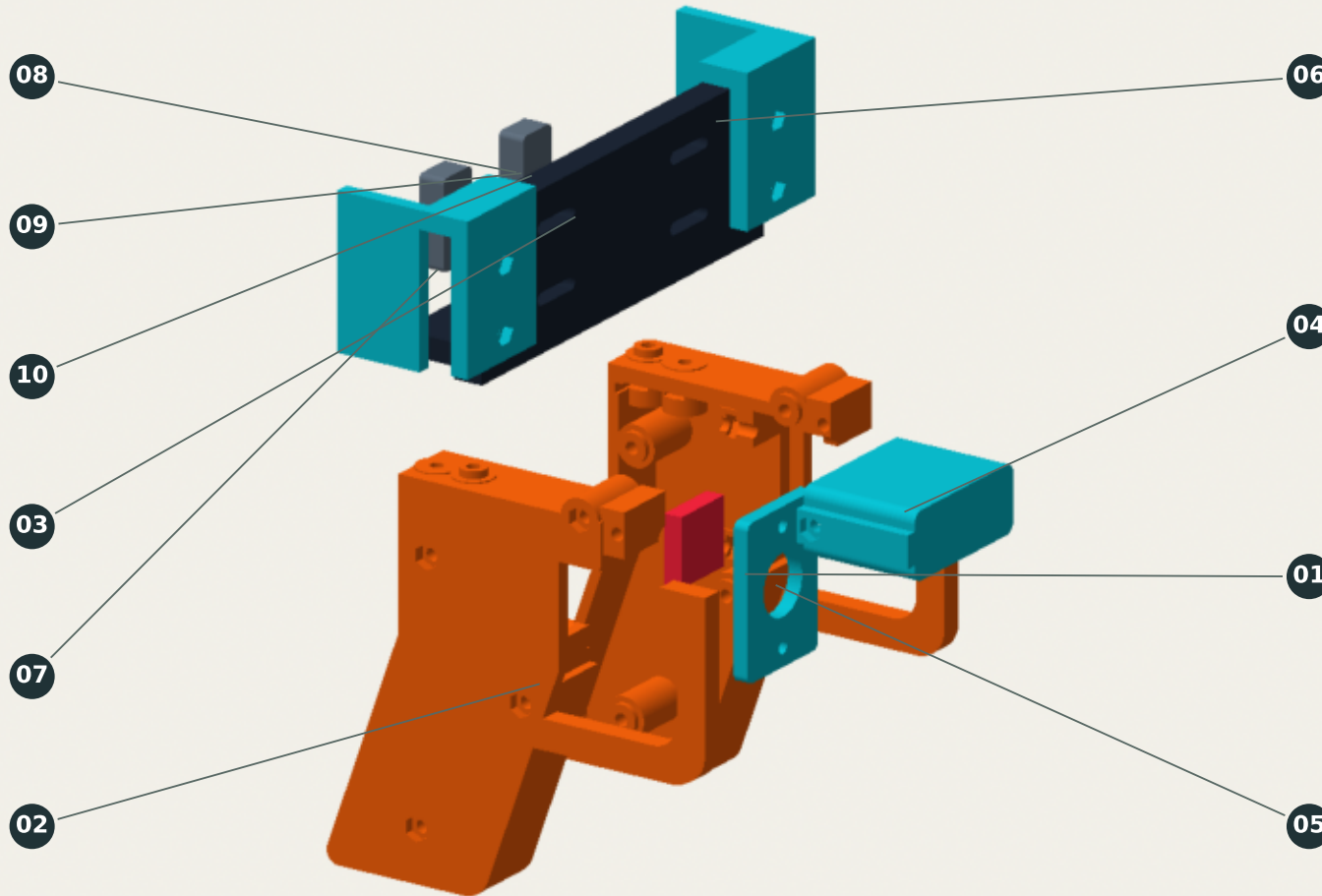


01 / Print separately. Assemble with fasteners.

Numbered printed components. The phone and purchased switch are separate items.



- 01 Left housing and grip**
Print quantity: 1
- 02 Right housing and grip**
Print quantity: 1
- 03 Universal phone dock**
Print quantity: 1
- 04 Closed arcade nose cap**
Print quantity: 1
- 05 Button trigger plate**
Print quantity: 1
- 06 Left sliding jaw**
Print quantity: 1
- 07 Right sliding jaw**
Print quantity: 1
- 08 Rigid shim 1 mm**
Print quantity: 2
- 09 Rigid shim 2 mm**
Print quantity: 2
- 10 Rigid shim 4 mm**
Print quantity: 2
- 11 Rigid shim 8 mm**
Print quantity: 2 (optional for this fit)
- 12 Joint and hardware fit coupon**
Print quantity: 1

02 / Individual print files

One STL per part type. Read dimensions as width × depth × build height; check supports in your slicer.

01 Left housing and grip



108.5 × 108.6 × 20.5 mm
Quantity 1

Exterior left face X = −19;
cavity up

02 Right housing and grip



108.5 × 108.6 × 19.0 mm
Quantity 1

Exterior right face X = 19;
cavity up

03 Universal phone dock



120.0 × 28.0 × 42.0 mm
Quantity 1

Shelf bottom Z = −4

04 Closed arcade nose cap



45.0 × 18.0 × 32.0 mm
Quantity 1

Closed front Y = 90; cavity
up

05 Button trigger plate



30.0 × 41.0 × 4.0 mm
Quantity 1

Flat rear Y = 42

06 Left sliding jaw



41.4 × 37.4 × 29.0 mm
Quantity 1

Exterior side wall

07 Right sliding jaw



41.4 × 37.4 × 29.0 mm
Quantity 1

Exterior side wall

08 Rigid shim 1 mm



12.0 × 24.0 × 1.0 mm
Quantity 2

Broad flat face

09 Rigid shim 2 mm



12.0 × 24.0 × 2.0 mm
Quantity 2

Broad flat face

10 Rigid shim 4 mm



12.0 × 24.0 × 4.0 mm
Quantity 2

Broad flat face

11 Rigid shim 8 mm



12.0 × 24.0 × 8.0 mm
Quantity 2 (optional for this fit)

Broad flat face

12 Joint and hardware fit coupon



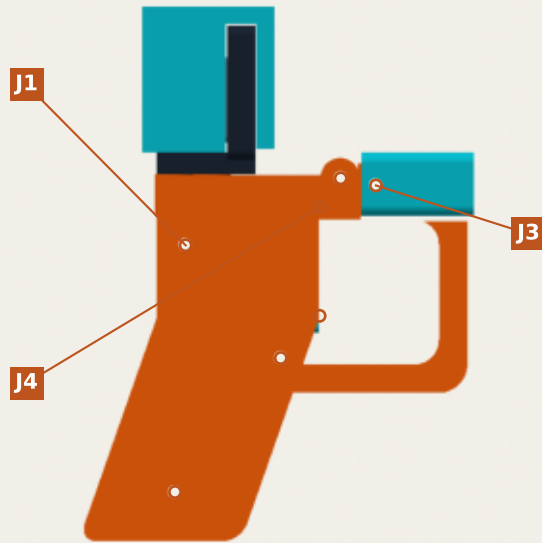
64.0 × 37.0 × 6.0 mm
Quantity 1

Flat base Z = 0

03 / Assembly geometry and joint references

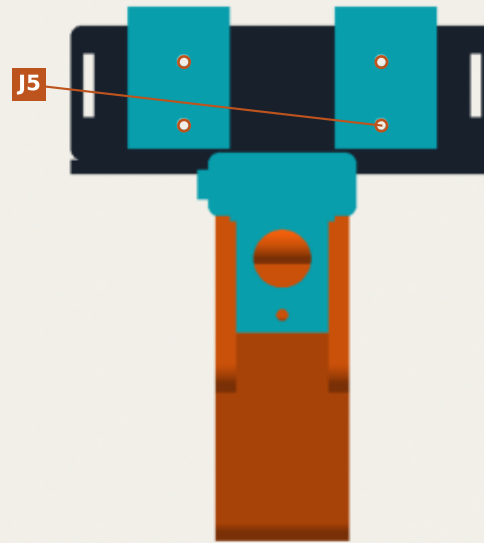
Overall plastic envelopes are derived from the exported meshes. See assembly-design.md for joint details.

SIDE VIEW · Y/Z



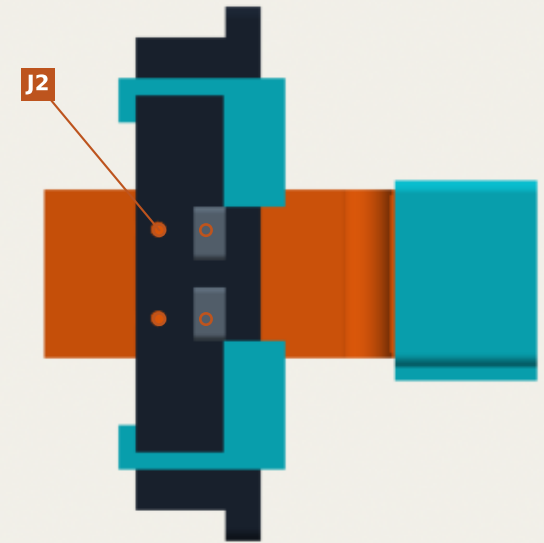
110.6 × 151.4 mm overall

FRONT VIEW · X/Z



120.0 × 151.4 mm overall

TOP VIEW · Y/X



110.6 × 120.0 mm overall

JOINTS TO CHECK BEFORE THE FULL PRINT

J1 / 4 × M3 × 40 mm — Close shell seam. Four annular locating lips. Thread ends project 2.3 mm beyond nuts.

J2 / 4 × M3 × 12 mm — Dock load path; recessed heads beneath removable liner.

J3 / 1 × M3 × 45 mm — Closed cap on receiver tongue; external side access.

J4 / 2 × M3 × 12 mm — Button plate, installed before nose. Seam-captured nuts.

J5 / 4 × M3 × 12 mm — Sliding jaws. Recessed front heads and rear captive nuts. Adjust with phone removed.

04 / Build order and first-print checks

Fit coupons first, then fasteners and switch, then the housing. Electronics are not supplied or implemented.

- 01** Print the fit coupon and button plate first. Trial the actual M3 hardware and locating fits, then the purchased switch in the plate. Correct individual CAD clearances if needed; do not scale the whole model.
- 02** Print the remaining parts in the supplied orientations. Clean mating faces and slicer-identified supports. Keep all v2 parts together; earlier v0/v1 parts are incompatible.
- 03** With the housing open, seat and retain FOUR J2 dock nuts and TWO J4 plate nuts. Mount the purchased switch in the loose plate with its own nut; confirm lead clearance and free travel. Keep electronics unpowered.
- 04** Close the shells on their locators using four J1 M3 × 40 bolts. Then attach the button plate with two J4 M3 × 12 bolts, approaching from the front BEFORE fitting the nose. Tighten gently by hand.
- 05** Fit the closed nose with one J3 M3 × 45 bolt and nut. Seat the dock on its pins and fit four J2 M3 × 12 bolts into the preloaded nuts, with phone and bottom liner removed for tool access.
- 06** Fit the jaws using four J5 M3 × 12 bolts and rear captive nuts. Adjust with the phone removed. Add two equal shim stacks, protective lining and the 15 mm restraint strap; follow the liner dimensions in the guide.
- 07** Bench-check with a phone-sized dummy, then the phone in its case with an independent tether. Verify retention, bolt ends, button travel, side controls, screen and all camera views before handling.

HARDWARE

4 × M3 × 40 mm button-head bolts — shell closure

1 × M3 × 45 mm button-head bolt — nose

10 × M3 × 12 mm button-head bolts — dock, jaws and plate

15 × standard M3 hex nuts. Modeled heads: Ø5.7 × 1.65 mm max; nuts: 5.5 mm across flats × 2.4 mm. No washers.

1 × 16 mm momentary pushbutton with its mounting nut

Soft protective lining, thin shim restraint tape, one 15 mm hook-and-loop strap, and an independent phone/case tether

BEFORE USING A PHONE

This is an untested mechanical prototype. Digital geometry does not establish phone retention, impact resistance or comfort. Inspect supports and layer direction in your actual slicer. Bluetooth, firmware, a battery mount and a physical game-firing demonstration are not included.